Power BI Classes 1-7: Study Guide

Formulas, findings and interview questions from the HR attrition project (IBM sample dataset, 1,470 employees)

The five steps of every Power BI project

Get data, clean it (Power Query), connect tables (model), calculate (DAX), visualise and share.

Class 1: Interface, loading data, first visuals

What you learned

- Three views: **Report** (charts), **Table** (see data), **Model** (relationships).
- Loaded the Excel/CSV file and built a card, bar chart, column chart and slicer.
- **Aggregation matters:** IDs (EmployeeNumber) use **Count**, never Sum. Amounts (income) use Sum or Average. Summing IDs gave a meaningless 2M.
- A card shows rounded numbers (1.47K); that is only display, the value is 1,470.

Q: What is the difference between a report and a dashboard?

A: A report has many pages and is built in Desktop. A dashboard is one page of pinned visuals, created in the Service.

Q: What is a slicer?

A: A clickable filter on the page. Selecting Male changes every visual on the page.

Class 2: Power Query (cleaning data)

What you learned

- Turn on **Column quality, distribution and profile**, and profile the **entire dataset** (not just the top 1000 rows).
- Checked: no blanks, no errors, no duplicates (EmployeeNumber 1,470 distinct and 1,470 unique).
- Removed constant columns: **EmployeeCount, Over18, StandardHours** (one value in every row).
- Created a **conditional column** Attrition Flag: if Attrition equals **Yes** then 1, else 0. Power Query is **case-sensitive**.
- Average of a 1/0 flag equals the rate: 237 / 1,470 = 16.1%.

Q: What is Power Query and what are Applied Steps?

A: It is the tool for cleaning data before it loads. Applied Steps record every transformation, so they repeat automatically on refresh.

Q: How do you check data quality?

A: Column quality and distribution on the full dataset: valid, error and empty percentages, distinct counts, data types and duplicates.

Q: Why create a 1/0 flag?

A: It lets an average give a rate, and it works easily inside measures.

Class 3: Merge and Append

What you learned

- Built a lookup table with **Enter data**: Education 1-5 = Below College, College, Bachelor, Master, Doctor.
- **Merged** it into the HR table with a **Left Outer** join, then expanded Education Level. Row count stayed **1,470**.
- **Append** stacks tables with the same columns (practised on small Sales 2024 and 2025 tables).

	Merge	Append
Combines	Side by side (adds columns)	On top of each other (adds rows)
Needs	A matching key column	Same column names
SQL equivalent	JOIN	UNION

Q: What can go wrong with a merge?

A: Duplicate keys in the lookup table multiply rows, so I always check the row count afterwards.

Q: Which join do you use most?

A: Left outer: keep every row of the main table and bring in matching details from the lookup.

Class 4: Data modelling

What you learned

- **Fact table** holds numbers and events (HR employees). **Dimension table** holds descriptions (Education_Lookup).
- **Star schema:** one fact table in the centre with dimension tables around it.
- Your relationship: **many-to-one** (many on the HR side, 1 on the lookup side) with a **single** filter direction. Filters flow from the 1 side to the many side.
- A slicer on the lookup table filtered the HR data (Master: 398 employees, 58 left, 14.6%).

Q: What is a star schema?

A: One central fact table linked to dimension tables, like a star. It keeps models fast and easy to understand.

Q: Why use a relationship instead of a merge?

A: The lookup is stored once, so the model stays smaller and cleaner. A merge repeats the text on every row.

Class 5: DAX basics (measures)

Formulas (create each with Home, then New measure)

```
Total Employees = COUNT('WA_Fn-UseC_-HR-Employee-Attrition'[EmployeeNumber])
Attrition Count = SUM('WA_Fn-UseC_-HR-Employee-Attrition'[Attrition flag])
Attrition Rate = DIVIDE([Attrition Count], [Total Employees])
Avg Monthly Income = AVERAGE('WA_Fn-UseC_-HR-Employee-Attrition'[MonthlyIncome])
```

- **Measure** is calculated when the visual is drawn and reacts to filters. **Calculated column** is calculated row by row at load and stored.
- **Count vs rate:** count = how many; rate = what share. R&D has the most leavers (133), but Sales has the highest rate (20.6%).

Q: Measure vs calculated column?

A: A calculated column is stored row by row when data loads. A measure is computed at query time from the active filters, so I use measures for totals and rates.

Q: Why DIVIDE instead of the / symbol?

A: DIVIDE handles division by zero without an error.

Class 6: CALCULATE and SWITCH

Formulas

```
Sales Attrition Count = CALCULATE([Attrition Count],
'WA_Fn-UseC_-HR-Employee-Attrition'[Department] = "Sales")
Overtime Attrition Rate = CALCULATE([Attrition Rate], 'WA_Fn-UseC_-HR-Employee-Attrition'[OverTime]
= "Yes")
No Overtime Attrition Rate = CALCULATE([Attrition Rate],
'WA_Fn-UseC_-HR-Employee-Attrition'[OverTime] = "No")
Age Group (calculated column) = SWITCH(TRUE(), 'WA_Fn-UseC_-HR-Employee-Attrition'[Age] < 30,
"18-29", 'WA_Fn-UseC_-HR-Employee-Attrition'[Age] < 40, "30-39",
'WA_Fn-UseC_-HR-Employee-Attrition'[Age] < 50, "40-49", "50+")
```

- **CALCULATE** evaluates a measure with a filter you choose, on top of the filters already active. Sales leavers = 92.
- **SWITCH(TRUE(), ...)** checks conditions in order and returns the first one that is true, like nested IFs but easier to read.
- In a chart, the axis field already filters each bar, so a chart can split Attrition Rate by OverTime without CALCULATE.

Q: What does CALCULATE do?

A: It changes the filter context and then evaluates an expression, for example attrition rate for overtime employees only.

Class 7: Building the dashboard

- Layout: title, four KPI cards, Gender and Education slicers, charts by department, age group, overtime and education, a department summary table and a key insights box.
- Check totals: the table total (1,470 / 237 / 16.1%) must match the KPI cards.
- Polish: light background, consistent card style, clean titles, highest bar highlighted, charts aligned. Export with File, Export, Export to PDF.

Q: Walk me through your dashboard.

A: I used the IBM HR attrition dataset with 1,470 employees. I cleaned it in Power Query, linked an education lookup, and wrote DAX measures for attrition rate and count. The dashboard shows KPIs and attrition by department, age, overtime and education. Overall attrition was 16.1%, Sales had the highest rate at 20.6%, and overtime employees left at about three times the rate of the rest.

All findings in one place

Cut	Result
Overall	1,470 employees, 237 left, 16.1% attrition
Department rate	Sales 20.6%, Human Resources 19.0%, R&D 13.8%
Department count of leavers	R&D 133, Sales 92, Human Resources 12
Overtime	Yes 30.5% vs No 10.4%
Age group	18-29 27.9%, 30-39 14.3%, 50+ 13.3%, 40-49 9.7%
Education level	Below College 18.2%, Bachelor 17.3%, College 15.6%, Master 14.6%, Doctor 10.4% (small group)
Gender	Female about 14.8%, Male about 17.0%

Quick revision: answer these out loud

- Why compare rates and not counts?
- What is the difference between merge and append?
- What is a star schema, and what do the 1 and * symbols mean?
- Measure vs calculated column, and why DIVIDE?
- What does CALCULATE do? Give an example from your project.
- Why did you check the row count after the merge?
- What are the limits of this analysis (sample data, small groups, correlation is not cause)?